

Skill Arena - Phase 4 Document 4

Bot Detection, Device Fingerprinting, Multi-Account Detection & Behavioral Security Architecture

Document Purpose

Define systems used to identify bots, detect multi-account abuse, analyze player behaviour and strengthen platform security.

Business Objective

Protect competitive integrity, prevent abuse and ensure all players compete fairly.

Security Philosophy

Trust behaviour, not claims. Every account, device and gameplay pattern is continuously evaluated.

Behavioral Security Engine

Central engine responsible for behavioural analysis and anomaly detection.

Bot Detection Framework

Detect scripted gameplay, automation tools, macros and non-human interactions.

Bot Detection Signals

Input timing, reaction patterns, completion consistency, navigation behaviour and challenge interactions.

Human Verification Models

Compare player activity against established human behavioural baselines.

Device Fingerprinting Engine

Generate device identities using hardware, software and environmental characteristics.

Fingerprint Components

Browser attributes, operating system, device configuration, network characteristics and security signals.

Trusted Device System

Maintain device reputation scores and trusted device histories.

Multi-Account Detection

Identify accounts controlled by the same individual or coordinated groups.

Detection Signals

Shared devices, shared payment methods, network similarities and behavioural overlaps.

Risk Correlation Engine

Links suspicious accounts, devices and activity patterns.

Behavioral Analytics

Monitor navigation habits, gameplay decisions, financial behaviour and account usage.

Anomaly Detection

Identify unusual actions, impossible patterns and suspicious deviations.

Account Risk Scoring

Generate dynamic trust and risk scores for each player.

Automated Enforcement

Warnings, verification requests, restrictions, reviews and account holds.

Manual Review Support

Investigators receive behavioural evidence and account correlation reports.

Gameplay Behaviour Monitoring

Track challenge completion patterns, movement behaviour and competitive activity.

Financial Behaviour Monitoring

Monitor deposits, withdrawals, reward usage and transaction trends.

Network Intelligence

Analyze IP usage, geographic changes and unusual access patterns.

Machine Learning Integration

Future support for adaptive detection and behavioural modelling.

Privacy Controls

Data collection governed by platform policies and legal requirements.

Audit Requirements

Risk decisions, detections and enforcement actions permanently logged.

Security Requirements

Behavioral data processed securely and protected against unauthorized access.

Performance Requirements

Detection systems operate continuously with minimal impact on gameplay.

Acceptance Criteria

Platform can identify suspicious automation, device abuse and coordinated account activity at scale.